Exercise 12

Chapter 2, Page 76

Introduction to Electrodynamics

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Solution

Verified

Step 1

Consider a uniformly charged solid sphere ρ.

Apply Gaussian surface with radius r , to fined electric field $\vec{E}$, so according to Gauss's law, we get.

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\varepsilon_0}$$

$$\vec{E}(4\pi r^2) = \frac{q_{in}}{\varepsilon_0}\ \hat{r}$$

$$\therefore\ \vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q_{in}}{r^2}\ \hat{r} \tag{1}$$

Step 2

Then, to find the value of q_{in}:

$$Q_{in} \to V \quad , \quad q_{in} \to v$$

whereas: V: the full volume

v: the volume of Gussian surface

$$\therefore \; q_{in} = \frac{Q_{in}}{V} v \quad , \quad \because \; \rho = \frac{Q_{in}}{V}$$

$$\therefore \; q_{in} = \rho v \quad , \quad v = \frac{4}{3}\pi r^3$$

$$\therefore \; q_{in} = \frac{4}{3}\rho\pi r^3 \qquad (2)$$

Step 3

By substituting (2) in (1), we get:

$$\vec{E} = \frac{1}{4\pi\varepsilon_0} \frac{\frac{4}{3}\rho\pi r^3}{r^2} \hat{r}$$

$$\therefore \; \vec{E} = \frac{\rho r}{3\varepsilon_0} \hat{r} \, , \quad \text{inside the volume}$$

Note that: this result agrees with the result of problem 2.8.

Result

$$\vec{E} = \frac{\rho r}{3\epsilon_0}\hat{\mathbf{r}}$$